Chapter 2 Study Guide: Linear Momentum

This guide covers the core concepts, key formulas, and problem-solving strategies for linear momentum, including both non-conserved and conserved systems.

1. Core Concepts

- **Linear Momentum ($\vec{P}$):** A vector quantity that measures an object's "quantity of motion."
  - **Formula:** $\vec{P} = m\vec{V}$
  - **Units:** $\mathrm{kg} \cdot \mathrm{m/s}$
  - Direction is the same as the velocity.
- **Momentum of a System ($\vec{P}_S$):** The vector sum of the individual momenta of all particles in the system.
  - **Formula:** $\vec{P}_S = \vec{P}_1 + \vec{P}_2 + \cdots = m_1\vec{V}_1 + m_2\vec{V}_2 + \ldots$
- **Center of Mass (G):** The momentum of a whole system can be treated as the momentum of its total mass moving at the velocity of its center of mass.
  - **Formula:** $\vec{P}_S = M_S \cdot \vec{V}_G$ (where M_S is the total mass)
  - *Key Insight:* If the total momentum $\vec{P}_S = \vec{0}$, the center of mass $\vec{V}_G$ is stationary (as seen in Example 1).

2. The Guiding Principle: Newton's 2nd Law

Everything in this chapter comes from one formula, the general form of Newton's 2nd Law:

$\frac{d\vec{P}_S}{dt} = \Sigma\vec{F}_{ext}$

This means: **The rate of change of a system's momentum is equal to the net external force acting on it.**

To solve any problem, you must first ask: **"Is the net external force $\Sigma\vec{F}_{ext}$ zero?"**

CASE 1: $\Sigma\vec{F}_{ext} \neq \vec{0} \implies$ NON-CONSERVATION OF MOMENTUM

If there is a net external force (like friction, braking, or gravity), momentum is NOT conserved.

We use the **Impulse-Momentum Theorem**:

- **Formula:** $\Delta\vec{P} = \Sigma\vec{F}_{ext} \cdot \Delta t \implies \vec{P}_{final} - \vec{P}_{initial} = \Sigma\vec{F}_{ext} \cdot \Delta t$
- **What it's for:** Problems involving a **force** acting over a **time** interval.
- **Examples:**
  - **E1 (Braking Vehicle):** Finding the **time** (t_1) it takes for friction/braking forces to stop a car.
  - **E2 (Inclined Plane):** Finding the friction force (f) by knowing the change in momentum over a known **time** (Δt).
  - **E3 (Projectile):** Finding the change in the y-momentum (P_y) over **time** (t) due to the external force of gravity.

Related Tool for Non-Conservative Forces:

- When a non-conservative force (like friction) acts over a **distance**, we use the **Work-Energy Theorem**.
- **Formula:** $\Delta ME = W_{nc} \implies (KE_f + GPE_f) - (KE_i + GPE_i) = -f \cdot D$
- **Examples:**
  - **E1 (Braking Vehicle):** Finding the stopping **distance** (D).
  - **E2 (Inclined Plane):** Finding the travel **distance** (D).

CASE 2: $\Sigma\vec{F}_{ext} = \vec{0} \implies$ CONSERVATION OF MOMENTUM

If there are no external forces (or they cancel out), momentum is conserved. This is typical for collisions, explosions, and jumps, where we analyze the system *during* the event.

We use the **Law of Conservation of Linear Momentum:**

- **Formula:** $\vec{P}_{initial} = \vec{P}_{final}$

We can break this into three types of problems:

A) Elastic Collisions (e.g., Application F1)

- **Definition:** Objects collide and bounce off each other without losing energy.
- **1. Momentum is Conserved:** $m_1\vec{V}_1 + m_2\vec{V}_2 = m_1\vec{V'}_1 + m_2\vec{V'}_2$
- **2. Kinetic Energy is Conserved:** $KE_{initial} = KE_{final}$

B) Perfectly Inelastic Collisions (e.g., Application F4)

- **Definition:** Objects collide and **stick together**.
- **1. Momentum is Conserved:** $m_1\vec{V}_1 + m_2\vec{V}_2 = (m_1 + m_2)\vec{V}_{syst}$
- **2. Kinetic Energy is LOST:** $KE_{final} < KE_{initial}$. (Energy is converted to heat, sound, etc.)

C) Explosions (e.g., Applications F2, F3)

- **Definition:** Objects in a system push each other apart. (The reverse of an inelastic collision).
- **1. Momentum is Conserved:** $\vec{P}_{initial} = \vec{P}_{final}$
  - (Often starts from rest: $\vec{0} = m_1\vec{V'}_1 + m_2\vec{V'}_2$)
- **2. Kinetic Energy is GAINED:** $KE_{final} > KE_{initial}$. (Energy is converted from potential energy, like a spring or chemical energy).

4. Summary Problem-Solving Strategy

1. **Define the System:** What objects are included?
2. **Analyze Forces:** Is $\Sigma\vec{F}_{ext} = \vec{0}$?
   - YES $\implies$ **Conserved:** Use $\vec{P}_i = \vec{P}_f$.
     - Is it elastic, inelastic, or an explosion? (Check KE).
   - NO $\implies$ **Not Conserved:**
     - Does the question involve **time**? Use $\Delta\vec{P} = \Sigma\vec{F}_{ext} \cdot \Delta t$.
     - Does the question involve **distance**? Use $\Delta ME = W_{nc}$.
3. **Set Coordinate System:** Define your positive ($\vec{i}, \vec{j}$) directions.
4. **Solve:** Substitute your values (with correct signs!) into the vector or scalar equations.